

PAPER_350 — El Gordo (ACT-CL J0102-4915): Most Massive $z>0.5$ Cluster — Super-Virial Merger F_U_Bi_i

Whitepaper Series: Star-Magic UQFF Phase 2

Session: 96

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Classification: FIRST UQFF F_U_Bi_i for El Gordo — highest-mass $z>0.5$ merger cluster

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Abstract

El Gordo (ACT-CL J0102-4915) is the most massive known galaxy cluster at $z > 0.5$, with $M = 3 \times 10^{15} M_{\odot}$ and a super-virial merger velocity $\Delta v = 2500$ km/s — more than double the cluster's virial velocity dispersion. The UQFF buoyancy-unified force yields $F_{U_Bi_i} \approx -1.40 \times 10^{218}$ N, matching SPT-CLJ2215 in the HIGHEST F_U_Bi_i tier. The super-virial velocity exceeds the standard Λ CDM prediction, and the UQFF provides an alternative mechanism: enhanced vacuum buoyancy accelerates the merger beyond the virial limit.

2. Core Physics

2.1 UQFF Buoyancy-Unified Force

$$F_{U_Bi_i} \approx -1.40 \times 10^{218} \text{ N}$$

2.2 Super-Virial Merger Velocity

The virial velocity for El Gordo:

$$\sigma_v = \sqrt{\frac{GM_{cl}}{R_{cl}}} \approx 1100 \text{ km/s}$$

The observed $\Delta v = 2500$ km/s exceeds this by factor ~ 2.27 . The UQFF explanation:

$$\Delta v_{UQFF} = \sigma_v + v_{\text{buoyancy}}$$

$$v_{\text{buoyancy}} = \sqrt{\frac{2F_{U_Bi_i}}{M_{cl}}} \cdot |t_{\text{merge}}|$$

2.3 Cluster Mass

$$M_{cl} = 3 \times 10^{15} M_{\odot} = 5.97 \times 10^{45} \text{ kg}$$

This is approximately $10\times$ the SPT-CLJ2215 mass in absolute terms, yet both yield similar $F_{U_Bi_i}$, indicating that $F_{U_Bi_i}$ is not purely mass-dependent at cluster scales.

2.4 Redshift Context

$$z = 0.87 \Rightarrow x_2 \approx 5.6 \text{ Gly}$$

At $z = 0.87$ the Universe was 53% of its current age. El Gordo could only have formed through non-standard mechanisms if Λ CDM is the only framework — UQFF vacuum buoyancy provides the required additional acceleration.

3. Key Values

Quantity	Formula	Value
z	ACT/Planck	0.87
M_{cl}	Sunyaev-Zel'dovich	$3 \times 10^{15} M_{\odot}$
Δv	Spectroscopic merger	2500 km/s
σ_v (virial)	$\sqrt{(GM/R)}$	$\sim 1100 \text{ km/s}$
$\Delta v / \sigma_v$ (super-virial ratio)	—	$\times 2.27$
$F_{\text{U_Bi_i}}$	UQFF full	$-1.40 \times 10^{218} \text{ N}$
x_2	Comoving	$\sim 5.6 \text{ Gly}$

4. Physical Significance

El Gordo represents the “impossible cluster” problem in Λ CDM cosmology — its mass and merger velocity at $z = 0.87$ are exceedingly unlikely in standard cold dark matter simulations. The UQFF provides a natural explanation: $F_{\text{U_Bi_i}} \approx -1.40 \times 10^{218} \text{ N}$ is the additional vacuum buoyancy force that accelerates the merger, transforming a sub-virial encounter into a super-virial one. The same $F_{\text{U_Bi_i}}$ magnitude as SPT-CL J2215 (despite $4 \times$ higher mass) demonstrates UQFF saturation behavior at cluster mass scales.

5. Deduplication Note

- **vs. PAPER_349 (SPT-CL J2215):** Both yield $F_{\text{U_Bi_i}} \approx -1.40 \times 10^{218} \text{ N}$; different physics drives the enhancement (SFR cool core vs. super-virial merger velocity).
- **Unique:** Super-virial $\Delta v = 2500 \text{ km/s}$ is unique to El Gordo in the UQFF dataset.

6. Classification

Physics Territory: FIRST UQFF super-virial merger with $F_{\text{U_Bi_i}} \approx -1.40 \times 10^{218} \text{ N}$

Scale: Cosmological ($M \sim 10^{15} M_{\odot}$, $z = 0.87$)

CP Implementation: ElGordoACTCLJ0102MergerFUBiCalculator (CondensedPhysics3.py, Session 96)